1. CB 4.54
2. CB 4.55
3. Consider the multivariate hierarchical model:

$$\begin{aligned} \mathbf{y} &= \mathbf{w} + \epsilon, \\ \mathbf{w} &\sim \text{Normal}\left(\mu \mathbf{1}, \mathbf{\Sigma}\right), \\ \epsilon &\sim \text{Normal}\left(\mathbf{0}, \tau^2 \mathbf{I}\right). \end{aligned}$$

   (a) What is the marginal distribution of $\mathbf{y}$?
   (b) Use the moment generating function to calculate the marginal mean of e^{y_i}, where y_i is an arbitrary element of $\mathbf{y}$? (Hint: Set $\mathbf{t}$ equal to a vector that is all 0s, except for the ith element which you set equal to 1.)
   (c) Consider two elements of $\mathbf{y}$, y_i and y_j. Use the moment generating function to calculate the marginal covariance between e^{y_i} and e^{y_j}.

4. Consider the discretized (or aggregated) log-Gaussian Cox process. For $i = 1, ..., n$,

$$\begin{aligned} y_i|\lambda_i &\overset{ind}{\sim} \text{Poisson}\left(\lambda_i\right), \\ \lambda_i &= e^{w_i}, \\ \mathbf{w} = \left(w_1, ..., w_n\right)^T &\sim \text{Normal}\left(\mathbf{0}, \mathbf{\Sigma}\right). \end{aligned}$$

   (a) What is the expected value of y_i? Use problem 3.
   (b) What is the covariance of y_i and y_j, $i \neq j$? Use problem 3.

5. CB 5.1
6. CB 5.2. For (a), consider Y to be the number of years until X_1 is exceeded. I find it helpful to find $P(Y = 1), P(Y = 2), ...$ and identify a pattern. Then, consider a u-substitution, where $u = F(x)$.
7. CB 5.6, this question should say (5.2.9) instead of (5.2.3).
8. CB 5.11
9. Let $X_1, \ldots, X_n$ be a random sample from a $N(\mu, \sigma^2)$ population. Calculate the variance of S^2, where

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2 \qquad \text{and} \qquad \overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

10. Let $U \sim N(0, 1)$, let $V \sim \chi^2_{(p)}$, and let U and V be independent. The book is pretty useful on this problem.

   (a) Find the joint pdf of

$$X = \frac{U}{\sqrt{V/p}} \qquad \text{and} \qquad Y = V.$$

(b) Find the marginal pdf of X and show that X follows a Student's t distribution with p degrees of freedom.

11. Let $U \sim \chi^2_{(p)}$, let $V \sim \chi^2_{(q)}$, and let U and V be independent.

    (a) Find the joint pdf of

    $$X = \frac{U/p}{V/q} \qquad \text{and} \qquad Y = U + V.$$

    (b) Find the marginal of X and show that X follows an $F_{p,q}$ distribution.

12. CB 5.16
13. CB 5.17 b,c,d (notice that part (a) is the same as problem 11. Don't do this again. For part (b), only find the mean.)
14. CB 5.18 (parts a and b only)
15. CB 5.20 (a only)